



Virtualisation

What is virtualisation?

Virtualisation is a technique by which hardware resources can be abstracted and divided, and these resources are allocated amongst multiple OS environments.

Why do we need virtualisation?

One of the many abilities that virtualisation provides is to run multiple OSes simultaneously on a single hardware platform, which is fast becoming an important solution to problems that confront Information Management today.

How does virtualisation work?

Virtualisation abstracts software from the underlying hardware. It cuts the link that ties a specific software stack to a particular server.

Where can I find an instance of virtualisation?

You could find a software implementation of virtualisation in any corporate computing environment, which needs to run and manage multiple OSes on a single server.

Who provides virtualisation solutions?

The best example of a software virtualisation provider is VMware Inc, a wholly-owned subsidiary of EMC Corporation. And in hardware virtualisation, one of the best examples is Intel, with its Intel Virtualization Technology.

When will Intel Virtualization Technology (IVT) ship?

Virtualisation is a long-term Intel initiative. Future generations of the technology are in design and will provide additional benefits. The first generation of IVT will ship in the second half of 2005 on new Pentium IV and Itanium II processors. In the first half of 2006, IVT will ship for Xeon processors and the Centrino mobile technology.

The SHARK keyboard is also compatible with natural English-language patterns. The average person has a writing vocabulary of about 10,000 words, so the patterns the computer will have to recognise remains somewhat contained.

SHARK works with a variety of keyboard types, and IBM is currently experimenting with a number of them. QWERTY keyboards, which were arranged to prevent mechanical problems caused by poorly configured keys, don't work that well. Alphabetically arranged keyboards are even worse, as the distance between some commonly associated letters is even farther apart.

Instead, IBM is conducting experiments with a modified version of its Atomik keyboard. This keyboard consists of three six-letter keys sandwiched by an upper and lower row of four letter keys. The keys are arranged to maximise letter associations.

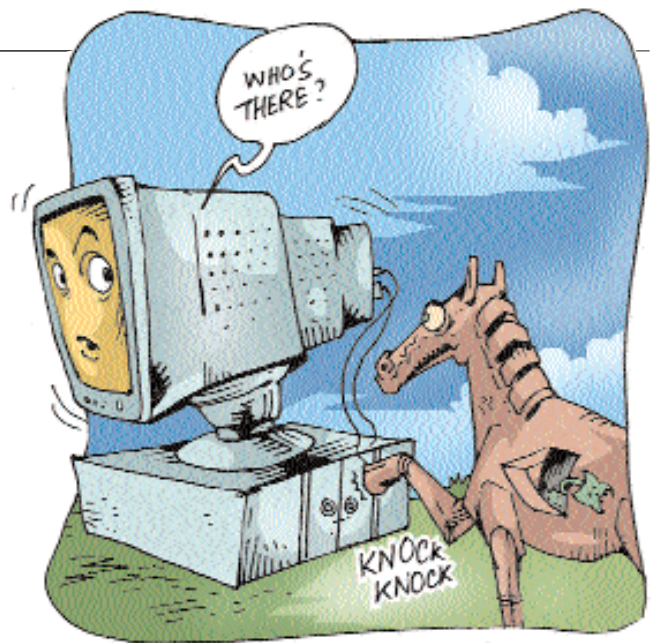
Beta copies of SHARK can be downloaded from IBM's AlphaWorks site, where it releases experimental software. Let's squiggle!

INCREASED FIREPOWER

Trojans Scouring The Net

The Net just got nastier. If recent reports by security software vendor Sophos are anything to go by, nearly 8,000 different pieces of malicious code have been detected so far this year—mainly coming from criminal gangs.

Sophos also reckons that the dramatic rise in the number of viruses, worms and Trojan horses this year can be attributed to organised criminals turning to cyber-crime. It picked up almost 60 per cent more malware in the first six months of this year than



the same time last year.

The biggest growth was in Trojans, programs that damage a user's files, steal information, and can even leave a backdoor for future attacks on the infected PC—a feature crooks value more than pure vandals. This, analysts say, is an indication that organised criminal gangs are now operating in the cyber world.

So if you were already worried about security on the Internet, this is as good a time as any to increase the alert to Code Orange! Time to hit 'Update Virus Definitions'.

SPEEDING UP

Optic Fibre On Your PCB

The speed of data transmission in your computer can now be spruced up through the PCB (Printed Circuit Board) itself, thanks to the use of fibre optics. A team of South Korean scientists claim that they have developed a new circuit board using fibre optics. This, according to them, will dramatically increase a computer's data transmission speed.

Park Hyo-Hoon, a professor at South Korea's

Information and Communications University in Daejeon, and Lee Yong-Tak, a professor of Gwangju Institute of Science and Technology, developed a new PCB wired by both optical fibres and traditional copper wires.

The introduction of dual-core processors by both Intel and AMD has increased processing speeds. However, lower data transmission speeds along the rest of the board due to bottlenecks has hampered the optimal productivity of these processors.

These bottlenecks are eliminated from the new PCB because of the use of optical fibres. With the new technology, it takes only one second to transmit a one-hour video clip of high-definition television while the existing technology takes tens of seconds, claimed the scientists.

They said data that needed high-speed transmission went through optical fibre wires while slow-speed data was directed through copper wires to prevent delays. The team said they expected the printed circuit board to come into commercial use around 2010 when production costs come down. ■

Compiled by Aditya Kuber,
Mithun Kidambi, Ram Mohan Rao
and Renuka Rane